
SIMULATION OF A $U(1)$ GAUGE THEORY ON A (1+1)-DIMENSIONAL LATTICE

IISER MOHALI SUMMER PROJECT REPORT

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ABSTRACT

The goal of this project is to simulate the Lattice Schwinger Model, a $U(1)$ gauge theory of Quantum Electrodynamics (QED) on a (1+1)-dimensional lattice. This is achieved using Hybrid Monte Carlo (Hamiltonian Monte Carlo) technique. We first discuss pseudo-random number generation, which is essential for Monte Carlo simulations. Then, we discuss computation of expectation values using simple Monte Carlo methods and later we generalise it for sampling from arbitrary probability distributions. Sampling random numbers from such probability distributions is found to be non-trivial and problematic. To overcome this, the Markov Chain Monte Carlo technique is introduced just after. The probability distribution function used in simulations of models based on path integral Monte Carlo is explained, and after that we explain an important mathematical concept used to simplify the equations obtained. We then discuss the imaginary time formalism in path integral Monte Carlo. We use Monte Carlo simulation of simple harmonic oscillator to show the working of all previously discussed concepts, and to explain the elementary method to discretise variables in the respective equations. The technique of Hybrid Monte Carlo (HMC) is discussed next, as it is one of the most efficient ways to simulate quantum fields on a lattice. After this, an overview of the continuum formulation of Quantum Electrodynamics (QED) in (3+1) dimensions is given as an application of the imaginary time formalism of the theory. We next discuss the procedural technique to regularize the theory on a lattice. We look at the theory dimensionally reduced to (1+1) dimensions, which is precisely the Lattice Schwinger model. Simulations using HMC are run in C++ and the observables are plotted to infer properties of this model.

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1 Pseudo Random Number Generation in C and C++

A pseudo-random number generator (PRNG) is an algorithm for generating a sequence of numbers whose properties approximate the properties of sequences of random numbers. The PRNG-generated sequence is not truly random, because it is completely determined by an initial value called the seed. To further closely replicate random numbers, this seed is taken as a random value from the hardware of a device. These generators are central to simulations.

1.1 The srand() and rand() functions

```
int rand (void);          void srand (unsigned int seed);
```

- The rand() function returns a pseudo-random integral number in the range between 0 and RAND_MAX, which is a constant defined in the header file <stdlib>.
- This number is generated by an algorithm which uses a seed (default seed = 1) to generate the series, which should be initialized to some distinctive value using function srand().
- In order to generate random-like numbers, srand() is usually initialized to some distinctive runtime value. This is distinctive enough for most trivial randomization needs. Eg. *srand(time(NULL))*.
- The formula is used to generate uniformly distributed random numbers in an open range (a, b) :

$$r = a + (b - a) * (rand() / RAND_MAX)$$

1.2 The ranlxd() and ranlxs() functions

- The ranlux generator is based on a dynamical system and has an astronomical period and is well suited for large-scale Monte Carlo simulations.
- The following function are used to generate n random double and floating point precision numbers and store them in an array.

```
void ranlxd(double r[], int n);    void ranlxs(float r[], int n);
```

To initialize the random number generators, the following functions are used.

```
void rlx_init(int level, int seed);    void rlx_init(int level, int seed);
```

2 Monte-Carlo Integration

In numerical integration, methods such as the trapezoidal rule use a deterministic approach. Monte Carlo integration, on the other hand, employs a non-deterministic approach: each realization provides a different outcome. In Monte Carlo, the final outcome is an approximation of the correct value with respective error bars, and the correct value is likely to be within those error bars. For the m -dimensional case:

$$I = \int_{\Omega} f(\bar{x}) d\bar{x}, \quad (2.1)$$

where Ω is a subset of R^m and has the following volume.

$$V = \int_{\Omega} d\bar{x}. \quad (2.2)$$

The Simplest Monte Carlo approach is to sample N points from a uniform random distribution on Ω .

$$\bar{x}_1, \dots, \bar{x}_N \in \Omega. \quad (2.3)$$

The Integral I can be approximated by

$$I \approx Q_N \equiv V \frac{1}{N} \sum_{i=1}^N f(\bar{\mathbf{x}}_i) = V \langle f \rangle. \quad (2.4)$$

The *Law of Large Numbers* ensures:

$$\lim_{N \rightarrow \infty} Q_N = I. \quad (2.5)$$

Given the estimation of I from Q_N , the error bars of Q_N can be estimated by the sample variance using the unbiased estimate of the variance

$$\text{Var}(f) \equiv \sigma_N^2 = \frac{1}{N-1} \sum_{i=1}^N (f(\bar{\mathbf{x}}_i) - \langle f \rangle)^2, \quad (2.6)$$

which leads to

$$\text{Var}(Q_N) = \frac{V^2}{N^2} \sum_{i=1}^N \text{Var}(f) = V^2 \frac{\text{Var}(f)}{N} = V^2 \frac{\sigma_N^2}{N}. \quad (2.7)$$

As long as all the σ_i are bounded $\text{Var}(Q_N)$ decreases asymptotically to zero as $\frac{1}{N}$. The estimation of the error of Q_N is thus,

$$\delta Q_N \approx \sqrt{\text{Var}(Q_N)} = V \frac{\sigma_N}{\sqrt{N}}. \quad (2.8)$$

This result does not depend on the number of dimensions of the integral, which is the promised advantage of Monte Carlo integration against most deterministic methods that depend exponentially on the number of dimensions.

2.1 Simple Monte Carlo for $\bar{\mathbf{x}}_i$ sampled from $p(\bar{\mathbf{x}})$

The Monte Carlo method can be extended to the case where the random numbers $\bar{\mathbf{x}}_i$ have to be sampled according to a normalized probability density function $p(\bar{\mathbf{x}})$.

The integral changes to,

$$I = \int_{\Omega} p(\bar{\mathbf{x}}) f(\bar{\mathbf{x}}) d\bar{\mathbf{x}} \approx \frac{1}{N} \sum_{i=1}^N f(\bar{\mathbf{x}}_i). \quad (2.9)$$

This is subject to the condition,

$$\int_{\Omega} p(\bar{\mathbf{x}}) d\bar{\mathbf{x}} = 1. \quad (2.10)$$

2.2 Importance Sampling

If the original $\bar{\mathbf{x}}_i$ are sampled from a distribution $p(\bar{\mathbf{x}})$, now suppose there is another normalized distribution $q(\bar{\mathbf{x}})$

$$I = \int_{\Omega} p(\bar{\mathbf{x}}) f(\bar{\mathbf{x}}) d\bar{\mathbf{x}} = \int_{\Omega} q(\bar{\mathbf{x}}) \left(\frac{p(\bar{\mathbf{x}})}{q(\bar{\mathbf{x}})} \right) f(\bar{\mathbf{x}}) d\bar{\mathbf{x}}. \quad (2.11)$$

Now considering $\left(\frac{p(\bar{\mathbf{x}})}{q(\bar{\mathbf{x}})} f(\bar{\mathbf{x}}) \right)$ as the new function and $q(\bar{\mathbf{x}})$ the new distribution to sample random numbers, we estimate the integral using Monte Carlo

$$I \approx \frac{1}{N} \sum_{i=1}^N \left(\frac{p(\bar{\mathbf{x}})}{q(\bar{\mathbf{x}})} f(\bar{\mathbf{x}}) \right), \quad (2.12)$$

$$\sigma^2(I) = \frac{1}{N} \sigma^2 \left(\frac{p(\bar{\mathbf{x}})}{q(\bar{\mathbf{x}})} f(\bar{\mathbf{x}}) \right),$$

$$\sigma^2(I) = 0,$$

for

$$q(\bar{\mathbf{x}}) = \frac{p(\bar{\mathbf{x}}) f(\bar{\mathbf{x}})}{\int_{\Omega} p(\bar{\mathbf{x}}) f(\bar{\mathbf{x}}) d\bar{\mathbf{x}}}. \quad (2.13)$$

The ideal choice hence is $q(\bar{x}) \propto |p(\bar{x})f(\bar{x})|$, but it requires a prior knowledge of the integral we wish to evaluate. Instead, if we use a $q(\bar{x})$ whose integral is known and whose behavior approximates that of $p(\bar{x})f(\bar{x})$, we can expect to reduce the variance.

Main problem with this method is the difficulty of generating random numbers with a probability density function $q(\bar{x})$, particularly in several dimensions.

3 Markov Chain Monte Carlo (MCMC)

Markov Chain Monte Carlo is also a random sampling method, and unlike the Simple Monte Carlo, the multi-dimensional is not sampled uniformly.

In MCMC the goal is to visit a point x with a probability proportional to some given distribution function $\pi(\bar{x})$. The distribution $\pi(\bar{x})$ is sampled using a Markov chain, which is a sequence of points $\bar{x}_1, \dots, \bar{x}_N$ that are locally correlated with the *Ergodic property*, which states that all points in the state space will be sampled given infinite time.

These points also have the *Markov property*, which states that the probability of moving to the next state depends only on the present state and not on the previous states.

We are interested in finding a stationary distribution, $\pi(\bar{x})$, starting from an initial distribution $\mu(\bar{x}_0)$. This is achieved through a series of Markov Chain jumps from $\bar{x}_i$ to $\bar{x}_{i+1}$ with the help of a transition probability function P .

A stationary distribution is defined by:

$$P\pi = \pi, \quad (3.1)$$

where π is a column vector. The equation intuitively states that any further jumps in the chain using P doesn't change the probability distribution.

3.1 Properties of Markov Chains:

3.1.1 Irreducibility

If S is the State Space, a Markov chain is irreducible if for all states $x, y \in S$, there exists a $t \geq 0$ such that $P^t(x, y) > 0$. Intuitively, it is the property which ensures that given any starting state, every other state in the state space can be achieved in finite time.

3.1.2 Aperiodicity

The property which leads to the occurrence of every state being aperiodic. Or in other words, no state in the state space can be periodic.

3.1.3 Reversibility

A Markov chain with stationary distribution $\pi(x)$ is called reversible if its transition matrix P exhibits *Detailed Balance*.

$$\pi(x)p(x, y) = \pi(y)p(y, x). \quad (3.2)$$

A reversible chain has the property that given we start with the stationary distribution the typical realization and then reversed chain have the same probabilistic behavior.

3.2 Convergence of Markov Chains

3.2.1 Uniqueness Theorem of Markov Chains

If a finite state Markov chain is *irreducible* and *aperiodic* then there exists a stationary distribution unique to that chain.

3.2.2 Theorem of Convergence in Finite Time

Given an *aperiodic* and *irreducible* Markov chain with state space $S = s_1, s_2, \dots, s_k$, and a transition matrix P , which has an arbitrary initial distribution μ . Then we have

$$\begin{aligned} P^n \mu &\rightarrow \pi \\ n &\geq 0. \end{aligned} \quad (3.3)$$

3.3 Metropolis-Hastings Algorithm

Suppose the desired stationary distribution of the MCMC has unnormalized density $\pi(x)$.

$$\tilde{\pi}(x) = \frac{\pi(x)}{Z} \quad (3.4)$$

for some positive constant Z .

If the conditional probability distribution $P(x_1|x_2)$ is not symmetric, then we use the Metropolis-Hastings algorithm to continue along the Markov chain. Proposal of a jump from the the current state x_1 to state x_2 is done by calculating the Metropolis-Hastings ratio, which is

$$r(x_1, x_2) = \frac{\pi(x_2)p(x_2, x_1)}{\pi(x_1)p(x_1, x_2)}. \quad (3.5)$$

Proposed jump is accepted with the probability

$$a(x_1, x_2) = \min(1, r(x_1, x_2)). \quad (3.6)$$

A special case of the Metropolis-Hastings is the Metropolis Algorithm, where $p(x, y) = p(y, x)$ and the ratio changes to the Metropolis ratio $r(x_1, x_2) = \frac{\pi(x_2)}{\pi(x_1)}$.

3.4 Thermalization in MCMC

Since we can start from any initial state in the state space, the chain takes some finite time-steps to concentrate around the states with high probability density. This process is called *Thermalization*.

These initial states in the chain are supposed to be discarded while computing the expectation values of observables.

4 Path Integral Monte Carlo

4.1 Principle of Least Action in Imaginary Time Formalism

Imaginary time τ may be obtained from real time t via a *Wick Rotation* by $\pi/2$ in the complex plane: $\tau = it$.

In physics, Wick rotation is a method of finding a solution to a mathematical problem in *Minkowski space* from a solution to a related problem in *Euclidean space* by means of a transformation that substitutes an imaginary-number variable for a real-number variable.

The Classical Action functional:

$$S[x(t)] = \int_t \left[\frac{m}{2} \left(\frac{dx(t)}{dt} \right)^2 - V(x(t)) \right] dt \quad (4.1)$$

such that the first order variation in the action

$$\delta S = 0 \quad (4.2)$$

Under Wick Rotation, the action becomes

$$S[x(-i\tau)] = \int_\tau \left[\frac{m}{2} \left(\frac{dx(-i\tau)}{d(-i\tau)} \right)^2 - V(x(-i\tau)) \right] d(-i\tau) \quad (4.3)$$

$$S[x] = i \int_\tau \left[\frac{m}{2} \left(\frac{dx}{d\tau} \right)^2 + V(x) \right] d\tau = iS',$$

such that

$$\delta S = \delta S' = 0. \quad (4.4)$$

4.2 Path Integral Approach in Imaginary Time Formalism

The Partition Function in Quantum Mechanics is ($\hbar$ is tuned to 1 throughout the model).

$$Z = \int_{\mathbf{x}(0)=x} \mathcal{D}\mathbf{x} e^{iS[\mathbf{x}, \dot{\mathbf{x}}]}, \quad (4.5)$$

and the wave function in the position representation becomes

$$\psi(x, t) = \frac{1}{Z} \int_{\mathbf{x}(0)=x} \mathcal{D}\mathbf{x} e^{iS[\mathbf{x}, \dot{\mathbf{x}}]} \psi_0(\mathbf{x}(t)), \quad (4.6)$$

where $\mathcal{D}\mathbf{x}$ denotes integration over all paths $\mathbf{x}$ with $\mathbf{x}(0) = x$ and S is the classical action. Taking Z inside the integral, we observe that $\frac{e^{iS}}{Z}$ acts like a probability weight term.

In Imaginary time formalism,

$$iS = - \int_{\tau} \left[m \left(\frac{dx}{d\tau} \right)^2 + V(x) \right] d\tau = -S', \quad (4.7)$$

and hence the probability weight term goes like $\frac{e^{-S'}}{Z}$ and is maximum where S' is minimum.

Each path becomes a state of the system and has an associated probability weight, and hence these states can be sampled from this probability distribution using the Metropolis-Hastings algorithm and Monte Carlo can be applied to find expectation values of observables.

5 The Simple Harmonic Oscillator using MCMC

5.1 Discretizing the Simple Harmonic Oscillator

We can write down the action of simple harmonic oscillator (in imaginary time formalism) in the following form:

$$S' = \int_{\tau_a}^{\tau_b} \left[\frac{m}{2} \left(\frac{dx}{d\tau} \right)^2 + \frac{m\omega^2 x^2}{2} \right] d\tau, \quad (5.1)$$

where $x(\tau)$ is position at time τ , m is the mass and ω is the frequency.

We discretize time on a one-dimensional lattice, with N sites, labelled by $i = 0, 1, \dots, N-1$, such that $\tau_a - \tau_b = Na$ where a is the *lattice spacing*. Also, $x_N = x_0$ when we impose periodic boundary condition on the 1-D lattice.

The discretized action becomes

$$S' = \frac{ma}{2} \sum_{j=0}^{N-1} \left[\left(\frac{x_{j+1} - x_j}{a} \right)^2 + \omega^2 \left(\frac{x_{j+1} + x_j}{2} \right)^2 \right] \quad (5.2)$$

Introducing dimensionless parameters

$$x_i = d_i \sqrt{\frac{a}{m}} \text{ and } \kappa = \frac{a^2 \omega^2}{4} \quad (5.3)$$

The Action now becomes

$$S' = \frac{(1 + \kappa)(d_0^2 + d_N^2)}{2} + (1 + \kappa) \left(\sum_{j=0}^{N-1} d_j^2 \right) - (1 - \kappa) \left(\sum_{j=0}^{N-1} d_j d_{j+1} \right) \quad (5.4)$$

The first term is a constant and can be set to zero without changing the physics. Performing another rescaling

$$u_i = d_i \sqrt{1 + \kappa} \text{ and } g = \frac{1 - \kappa}{1 + \kappa} \quad (5.5)$$

The discretized action finally takes the form

$$S' = \left(\sum_{j=0}^{N-1} u_j^2 \right) - g \left(\sum_{j=0}^{N-1} u_j u_{j+1} \right) \quad (5.6)$$

5.2 Monte Carlo and the Metropolis Update

The x_i are updated independently by proposing a random shift of δ' such that, $-\Delta < \delta' < \Delta$, from a uniform probability distribution. The change in action then becomes,

$$\delta S' = \delta'[\delta' + 2u_j - g(u_{j-1} + u_{j+1})] \quad (5.7)$$

$$u_j^{new} = u_j + \delta' \quad (5.8)$$

is accepted with the probability

$$\min(1, e^{-\delta S'}). \quad (5.9)$$

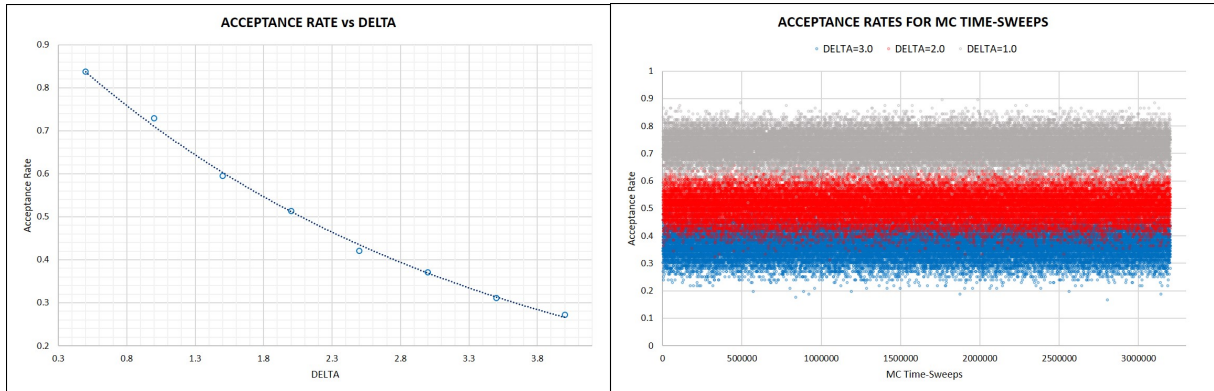
5.3 Simulation Results

Simulation parameters:

- Number of time slices $N = 32$.
- Number of Monte Carlo sweeps = 10^6
- Δ varied for observing the trend in acceptance rates
- $a\omega = 0.25$
- $\kappa = \frac{1}{4}(a\omega)^2 = 0.0156$
- $g = \frac{1-\kappa}{1+\kappa} = 0.9692$

5.3.1 Acceptance Rate

Δ is a simulation parameter whose value can be arbitrarily fixed to constrain the acceptance rate to be $\sim 50\%$. Lower acceptance rate implies more time wasted on rejections and a higher acceptance rate implies time wasted due to slow movement through the phase space.



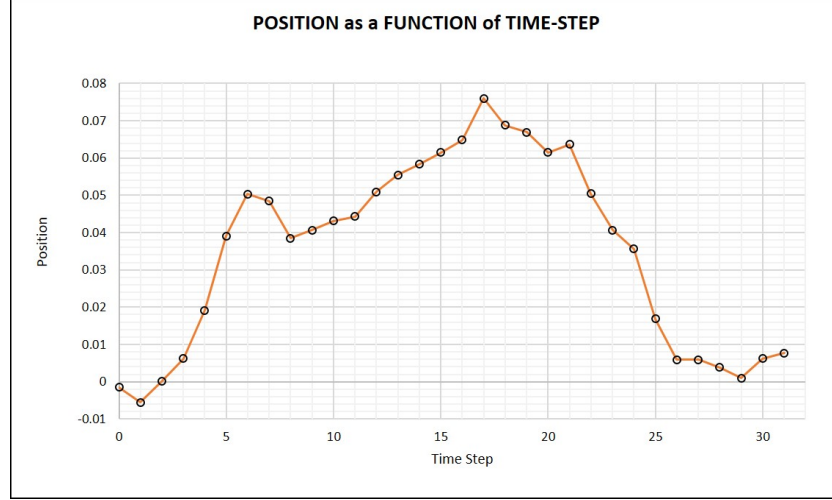
(a) Acceptance rate against Δ .

(b) Monte Carlo time history of acceptance rate for various values of Δ .

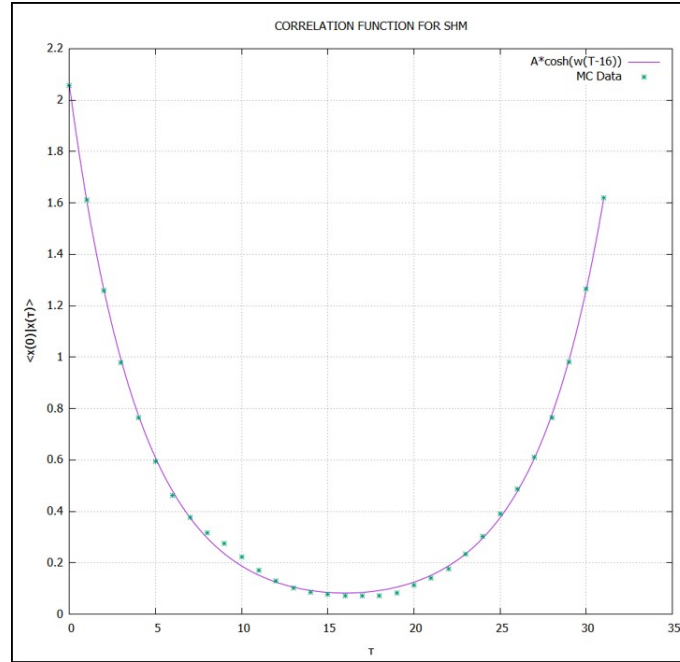
Figure 5.1: Acceptance rates for various values of Δ .

5.3.2 Positions at all Time Steps

The ideal graph should be sinusoidal. The observed graph is approaching a sinusoid. The reason for the deviation from ideality is smaller number of MC sweeps taken.

Figure 5.2: Position variable $u[t_i]$ as a function of time step t_i

5.3.3 Correlation Function

Figure 5.3: The Correlation Function $\langle x(\tau)x(0) \rangle$ against time τ .

The correlation function is periodic due to the periodic boundary condition imposed on the lattice. The correlated data is fit to a functional form $A * \cosh(\omega(\tau - N/2))$, with $A = 0.0819$, $\omega = 0.2447$ and $N = 32$.

6 Hamiltonian Monte Carlo

For Hamiltonian Monte Carlo, we usually use Hamiltonian functions that can be written as

$$\mathcal{H}(p, q) = K(p) + U(q), \quad (6.1)$$

where $U(q)$ is the potential energy and will be defined to be minus the log probability density of the distribution for q , the variable that needs to be sampled. $K(p)$ is called the kinetic energy and is usually defined as

$$K(p) = \frac{p^T M^{-1} p}{2}, \quad (6.2)$$

where M is a symmetric, positive definite *mass matrix*, which is typically diagonal. This form corresponds to minus the log probability density of the zero-mean Gaussian distribution with covariance matrix M . The time evolution of q and p follow Hamilton's equations.

6.1 Properties of Hamiltonian Dynamics

Several properties of Hamiltonian dynamics are crucial to its use in constructing MCMC updates.

6.1.1 Reversibility

The mapping T_s from the state $(q(t), p(t))$ to the state $(q(t+s), p(t+s))$ is one-one, and hence has an inverse T_{-s} obtained by negating the time derivatives in Hamilton's equations. Hence the Markov Chain formed using Hamilton's dynamics is also reversible, which is desirable.

6.1.2 Conservation of the Hamiltonian

For Metropolis updates using a proposal found by Hamiltonian dynamics, the acceptance probability is 1 if $\mathcal{H}$ is kept invariant, and close to 1 when the errors are accounted for.

6.1.3 Volume Preservation

From *Liouville's Theorem*, the volume in (q, p) is conserved. The significance for MCMC is change in volume is not needed to be accounted for in the acceptance probability for Metropolis updates.

6.2 Discretizing Hamilton's Equations: The Leap Frog Algorithm

Hamilton's equations in the continuum form are

$$\frac{dp_i}{dt} = -\frac{\partial \mathcal{H}}{\partial q_i} \quad \text{and} \quad \frac{dq_i}{dt} = \frac{\partial \mathcal{H}}{\partial p_i}, \quad (6.3)$$

with ϵ as a simulation parameter, the Hamilton's equations in the Leap Frog algorithm take the form

$$\begin{aligned} p_i(t + \frac{\epsilon}{2}) &= p_i(t) - \frac{\epsilon}{2} \frac{\partial \mathcal{H}}{\partial q_i} \Big|_{q(t)}, \\ q_i(t + \epsilon) &= q_i(t) + \epsilon \frac{\partial \mathcal{H}}{\partial p_i} \Big|_{p(t+\frac{\epsilon}{2})}, \\ p_i(t + \epsilon) &= p_i(t) - \frac{\epsilon}{2} \frac{\partial \mathcal{H}}{\partial q_i} \Big|_{q(t+\epsilon)}. \end{aligned} \quad (6.4)$$

6.3 The HMC Algorithm

The distribution we wish to sample can be related to a potential energy function via the concept of a *canonical distribution* from statistical mechanics. The Hamiltonian is an energy function and defines a joint distribution for (q, p)

$$P(q, p) = \frac{1}{Z} \exp(-\mathcal{H}(q, p)) = \frac{1}{Z} \exp(-U(q)) \exp(-K(p)). \quad (6.5)$$

Because q and p are independent variables, $U(q)$ and $K(p)$ act as separate energy functions and leading to separate canonical distributions. q_i represent the variables of interest, p_i are introduced just to allow Hamiltonian dynamics to operate.

Since $K(p)$ has a quadratic form, the canonical distribution for p is simply a Gaussian, and hence it is sampled from one. This is where the interpretation of $U(q)$ as the negative logarithm of the required probability distribution comes from.

7 U(1) Gauge Theory on a Lattice

7.1 Continuum Considerations

The action for renormalizable Quantum Electrodynamics (QED) is

$$S_{QED} = \int d^4x \left[-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \bar{\psi}(i\gamma^\mu \partial_\mu - M)\psi - e\bar{\psi}\gamma_\mu\psi A^\mu \right]. \quad (7.1)$$

The γ^μ are the Dirac gamma matrices which appear in theories containing a spin-1/2 field.

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \quad \text{where } \eta^{\mu\nu} = \text{diag}(1, -1, -1, -1) \text{ is the metric.} \quad (7.2)$$

The electromagnetic field is given by the $U(1)$ gauge vector field A^μ with field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ while the fermion (electron) field is given by the spinor field ψ with mass M . The interaction term is proportional to the electric charge e given by the last term $-e\bar{\psi}\gamma_\mu\psi A^\mu$.

The Euler-Lagrange equations derived from S_{QED} are the Maxwell equations and the Dirac equation

$$\begin{aligned} \partial_\mu F^{\mu\nu} &= j^\nu = e\bar{\psi}\gamma^\nu\psi, \\ (i\gamma^\mu \partial_\mu - m - e\gamma_\mu A^\mu)\psi &= 0. \end{aligned} \quad (7.3)$$

The above action is also invariant under $U(1)$ gauge transformations

$$\begin{aligned} A_\mu &\rightarrow A_\mu + \partial_\mu X, \\ \psi &\rightarrow \exp(-ieX) \psi, \\ \bar{\psi} &\rightarrow \bar{\psi} \exp(ieX). \end{aligned} \quad (7.4)$$

The Feynman path integral is

$$Z = \int \mathcal{D}A^\mu \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp(iS_{QED}) \quad (7.5)$$

The QED action can be separated into a fermionic part and a gauge field part.

The fermionic action is

$$S_F = \int d^4x \left[\bar{\psi}(i\gamma^\mu \partial_\mu - M)\psi - e\bar{\psi}\gamma_\mu\psi A^\mu \right] = \int d^4x \left[\bar{\psi}(i\gamma^\mu D_\mu - M)\psi \right], \quad (7.6)$$

where $D_\mu = \partial_\mu + ieA_\mu$ is the covariant derivative.

The gauge field action is

$$S_G = \int d^4x \left[-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right]. \quad (7.7)$$

Now, rotating S_{QED} to Euclidean signature in order to convert the theory into a statistical field theory.

This is done by

- $x_0 \rightarrow -i x_4$,
- $\psi^E(x) = \psi(\vec{x}, x_4)$ for $\psi(x) = \psi(\vec{x}, t)$,
- $A_0 \rightarrow -iA_4$ and $D_0 \rightarrow -iD_4$,
- $\gamma_4^E = \gamma^0, \gamma_k^E = i\gamma^k$,
- Multiplying the obtained expressions for actions by i .

The Euclidean fermionic action is

$$S_F^{Eucl} = \int d^4x \left[\bar{\psi}^E (i\gamma_\mu^E D_\mu + M)\psi^E \right]. \quad (7.8)$$

The Euclidean gauge field action is

$$S_G^{Eucl} = \int d^4x \left[\frac{1}{4} F_{\mu\nu}^2 \right]. \quad (7.9)$$

The full Euclidean action is

$$S_{QED}^{Eucl} = S_G^{Eucl} + S_F^{Eucl}. \quad (7.10)$$

7.2 Lattice Regularization

7.2.1 Lattice fermion fields

A regularization which maintains exact chiral invariance of the above action can be achieved by replacing the Euclidean four-dimensional spacetime by a four dimensional hypercubic lattice of N^4 sites. Every point on the lattice is specified by 4 integers ($0 < n_i < N$), which we denote collectively by $n = (n_1, n_2, n_3, n_4)$ where n_4 denotes Euclidean time. The lattice is assumed to be periodic. Thus $x_\mu = an_\mu$ where a is the infra-red cutoff or the lattice spacing and $L = aN$ is the ultra-violet cutoff or the linear size of the lattice. At each site we associate a spinor variable $\psi(x)$ and the derivative is replaced by

$$\partial_\mu \rightarrow \frac{1}{2a} \left[\psi(n + \hat{\mu}) \right]. \quad (7.11)$$

The vector $\hat{\mu}$ is the unit vector in the μ direction. With this prescription, the action becomes

$$S_F = \sum_n \sum_m \sum_\alpha \sum_\beta \bar{\psi}_\alpha(n) K_{\alpha\beta}(n, m) \hat{\psi}_\beta(m), \quad (7.12)$$

where

$$K_{\alpha\beta} = \frac{1}{2} \sum_\mu (\gamma_\mu)_{\alpha\beta} \left(\delta_{m, n+\mu} - \delta_{m, n-\mu} \right) + \hat{M} \delta_{\alpha\beta} \delta_{mn}. \quad (7.13)$$

This action has the following global $U(1)$ symmetry

$$\hat{\psi}_\alpha(n) \rightarrow G \hat{\psi}_\alpha(n), \quad \bar{\hat{\psi}}_\alpha(n) \rightarrow \bar{\hat{\psi}}_\alpha(n) G^{-1}. \quad (7.14)$$

The phase $G = \exp(-iX)$ is an element of $U(1)$. By requiring the theory to be invariant under local $U(1)$ symmetry, we arise at a problem, which is how to make the bilinear fermionic terms in the above action gauge invariant.

In the continuum formulation, under a local $U(1)$ transformation, the bilinear term transforms as

$$\bar{\psi}(x) \psi(y) \rightarrow \bar{\psi}(x) G^{-1}(x) G(y) \psi(y). \quad (7.15)$$

This bilinear can be made gauge covariant by inserting the Schwinger line integral

$$U(x, y) = \exp \left(ie \int_x^y dz_\mu A_\mu(z) \right), \quad (7.16)$$

which transforms as

$$U(x, y) = G(x) U(x, y) G^{-1}(y). \quad (7.17)$$

Therefore the fermionic bilinear

$$\bar{\psi}(x) U(x, y) \psi(y) = \bar{\psi}(x) \exp \left(ie \int_x^y dz_\mu A_\mu(z) \right) \psi(y), \quad (7.18)$$

is $U(1)$ gauge invariant. For $y = x + \epsilon$ we have,

$$U(x, x + \epsilon) = \exp(ie\epsilon_\mu A_\mu). \quad (7.19)$$

To get local $U(1)$ gauge invariance the bilinear terms are changed accordingly and then we obtain the action

$$S_F = \hat{M} \sum_n \sum_\alpha \bar{\hat{\psi}}_\alpha(n) \hat{\psi}_\alpha(n) - \frac{1}{2} \sum_n \sum_\alpha \sum_\beta \sum_\mu \left[(\gamma_\mu)_{\alpha\beta} \bar{\hat{\psi}}_\alpha(n + \hat{\mu}) U_{n+\hat{\mu}, n} \hat{\psi}_\beta(n) - (\gamma_\mu)_{\alpha\beta} \bar{\hat{\psi}}_\alpha(n) U_{n, n+\hat{\mu}} \hat{\psi}_\beta(n + \hat{\mu}) \right]. \quad (7.20)$$

The $U(1)$ element $U_{n,n+\hat{\mu}}$ lives on the lattice link connecting the two points n and $n + \hat{\mu}$. The link variable is explicitly given by

$$\begin{aligned} U_{n,n+\hat{\mu}} &= \exp(i\phi_\mu(n)) \equiv U_\mu(n) \\ U_{n+\hat{\mu},n} &= \exp(-i\phi_\mu(n)) \equiv U_\mu^\dagger(n) \\ \phi_\mu(n) &= eaA_\mu(n) \in [0, 2\pi]. \end{aligned} \quad (7.21)$$

7.2.2 Lattice gauge fields

The simplest non-trivial loop in the lattice is called the Plaquette. In the $\mu - \nu$ plane, we have

$$\begin{aligned} U_P &\equiv U_{\mu\nu}(n) = U_\mu(n)U_\nu(n+\hat{\mu})U_\mu^\dagger(n+\hat{\mu})U_\nu^\dagger(n) \\ &= \exp(i\epsilon a^2 F_{\mu\nu}(n)), \end{aligned} \quad (7.22)$$

where

$$F_{\mu\nu} = \frac{1}{a} \left[A_\nu(n+\hat{\mu}) - A_\nu(n) - A_\mu(n+\hat{\nu}) + A_\mu(n) \right]. \quad (7.23)$$

So, in the continuum limit, which is $a \rightarrow 0$, the following function approaches the continuum gauge action

$$\frac{1}{e^2} \sum_n \sum_{\mu,\nu} \left[1 - \frac{1}{2} (U_{\mu\nu} + U_{\mu\nu}^\dagger) \right] = \frac{a^4}{4} F_{\mu\nu}^2. \quad (7.24)$$

The $U(1)$ gauge action on a lattice is therefore

$$S_G = \frac{1}{e^2} \sum_P \text{Re} \left[1 - U_P \right]. \quad (7.25)$$

8 The Lattice Schwinger Model

In physics, the Schwinger model, named after Julian Schwinger, is a model describing QED in $(1+1)$ dimensions with a Dirac fermion and is a toy model for Quantum Chromodynamics (QCD). Restricting μ to $\{1, 2\}$ in the expressions for Euclidean action formulated above leads to the equations of the Schwinger Model. This model can be solved exactly and is used as a toy model for other more complex theories.

8.1 Putting Fermions on the Lattice for Simulations

Since the fermions follow Grassmann algebra and not regular algebra, they cannot directly simulated on a lattice.

The QED path integral with U as the Link Fields, and ψ and $\bar{\psi}$ as the fermion fields

$$\begin{aligned} Z_{QED} &= \left[\int \mathcal{D}U \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp(-S_G[U] - \bar{\psi} D[U] \psi) \right] \\ &\propto \left[\mathcal{D}U \det(D[U]) \exp(-S_G[U]) \right] \end{aligned} \quad (8.1)$$

The determinant can be represented by psuedo-fermion fields ϕ

$$\det(D) \propto \int \mathcal{D}\phi^\dagger \mathcal{D}\phi \exp(-\phi^\dagger D^{-1} \phi). \quad (8.2)$$

8.2 Expression for the action

S_{QED} hence becomes,

$$S_{QED} = \beta \sum_P \text{Re} \left[1 - U_P \right] + \phi^\dagger \frac{1}{MM^\dagger} \phi. \quad (8.3)$$

The first term is the gauge field action S_G and the second term is the fermionic action S_F . $\beta = \frac{1}{e^2}$ is the coupling constant and M is the Wilson Dirac operator.

$$S_G = \beta \left(1 - \cos \left(A_\mu(n) + A_\nu(n + \hat{\mu}) - A_\mu(n + \hat{\nu}) - A_\nu(n) \right) \right) \quad (8.4)$$

$$M_{n\alpha, m\beta} = (m_0 + 2r) \delta_{nm} \delta_{\alpha\beta} - \frac{1}{2} \sum_{\mu} \left[(r - \gamma_\mu)_{\alpha\beta} U_{n,\mu} \delta_{n, m - \hat{\mu}} + (r + \gamma_\mu)_{\alpha\beta} U_{n,\mu}^\dagger \delta_{n, m + \hat{\mu}} \right] \quad (8.5)$$

This looks very similar to the expression for lattice S_F^{Eucl} in the previous section (Eq. 7.20). The only difference is a new mass-like term r is introduced. This is to account for the *Fermion doubling problem*, which causes an overcounting of the fermions on the lattice.

In 2 dimensions, the Dirac γ matrices are the Pauli matrices.

$$\gamma_1 = \sigma_1, \gamma_2 = \sigma_2, \gamma_5 = \sigma_3. \quad (8.6)$$

M is γ_5 hermitian and $MM^\dagger$ is a positive definite. ($Q = M\gamma_5$)

$$\begin{aligned} M^\dagger &= \gamma_5 M \gamma_5 \\ MM^\dagger &= Q^2. \end{aligned} \quad (8.7)$$

Since M has these properties, we define a new quantity R such that,

$$R = Q^{-1} \phi. \quad (8.8)$$

The fermionic part of the action now becomes

$$S_F = \phi^\dagger \frac{1}{MM^\dagger} \phi = \phi^\dagger \frac{1}{Q^2} \phi = R^\dagger R. \quad (8.9)$$

This quadratic term in the action leads to R being directly sampled from a Gaussian distribution. ϕ can be reproduced by applying Q on R .

8.3 Implementing the HMC Algorithm

Let the gauge fields be $A_{n,\mu}$, which is shorthand for the μ^{th} component of the field at site n . Since the gauge fields need to be sampled from a probability distribution which goes like $e^{-S_{QED}}$ (as we inferred from the path integral formulation in imaginary time), momentum variables conjugate to these gauge fields are introduced and the Hamiltonian is written as

$$\mathcal{H} = \left(\frac{1}{2} \sum_{n,\mu} p_{n,\mu}^2 \right) + S_{QED}. \quad (8.10)$$

$p_{n,\mu}$ are sampled randomly from a Gaussian distribution $P \sim \exp(-\frac{p^2}{2})$ and the Hamiltonian is computed.

The equations for evolution of the gauge fields and the conjugate momenta become

$$\frac{dA_{n,\mu}}{dt} = p_{n,\mu}, \quad (8.11)$$

$$\begin{aligned} \frac{dp_{n,\mu}}{dt} &= -\frac{\partial S_{QED}}{\partial A_{n,\mu}} = -\frac{\partial S_G}{\partial A_{n,\mu}} + \eta^\dagger \frac{\partial(Q^2)}{\partial A_{n,\mu}} \eta, \\ \eta &\equiv \frac{1}{Q^2} \phi. \end{aligned} \quad (8.12)$$

The second term is written in such a way because variation of inverse matrices go as follows

$$\delta(X^{-1}) = -X^{-1} \delta(X) X^{-1}. \quad (8.13)$$

The derivative of S_{QED} with respect to $A_{n,\mu}$ computes to

$$\begin{aligned} \frac{\partial S_{QED}}{\partial A_{n,\mu}} = & \beta \left(\sin \left(A_\mu(n) + A_\nu(n + \hat{\mu}) - A_\mu(n + \hat{\nu}) - A_\nu(n) \right), \right. \\ & \left. - \sin \left(A_\mu(n - \hat{\nu}) + A_\nu(n + \hat{\mu} - \hat{\nu}) - A_\mu(n) - A_\nu(n - \hat{\nu}) \right) \right). \end{aligned} \quad (8.14)$$

The first term inside the bracket is from simple differentiation, but the second term is a little non-trivial. It appears by passive shifting the lattice points by one unit in $-\hat{\nu}$ direction to compute the term $\frac{\partial A_\mu(n + \hat{\nu})}{\partial A_{n,\mu}}$.

These updates are carried out via the Leapfrog algorithm and each update is accepted according to the probability

$$P_A(\mathcal{H} \rightarrow \mathcal{H}') = \min(1, \exp(\mathcal{H} - \mathcal{H}')), \quad (8.15)$$

where

$$\begin{aligned} \mathcal{H}(A_{n,\mu}, p_{n,\mu}) &= \sum_{n,\mu} \frac{p_{n,\mu}^2}{2} + S_G(A_{n,\mu}) + R^\dagger R, \\ &\text{and} \\ \mathcal{H}'(A'_{n,\mu}, p'_{n,\mu}) &= \sum_{n,\mu} \frac{p'^2_{n,\mu}}{2} + S_G(A'_{n,\mu}) + R'^\dagger R'. \end{aligned} \quad (8.16)$$

The conservation of the Hamiltonian (considering simulation errors) guarantees large acceptance rates.

8.4 Conjugate Gradient Algorithm

It is an algorithm to solve the matrix equation $\hat{F}p = q$ for p , where $\hat{F}$ is a Hermitian positive definite matrix and p and q are column vectors. It is based on the principle that solving the above linear equation is the same as extremizing a corresponding quadratic function: $g(p^\dagger) = \frac{p^\dagger \hat{A} p}{2} - p^\dagger q$ w.r.t $p^\dagger$.

The steps of the algorithm are:

1. Set the initial guess for p as x
2. Calculate the residue $r = y - \hat{A}x$
3. $norm_sq = r^\dagger r$
4. Assign the variable $p = r$
5. Set $\alpha = \frac{r^\dagger r}{p^\dagger \hat{A}^2 p}$
6. Update guess $x \rightarrow x + \alpha p$
7. Update residue $r \rightarrow r - \alpha \hat{A}^2 p$
8. $err_sq = r^\dagger r$
9. Check if $err_sq < \epsilon^2$, where ϵ is the required precision.
 - Yes: Assign $p = x$. This is the solution.
 - No: Set $\beta = \frac{err_sq}{norm_sq}$, update $p \rightarrow \beta p + r$ and repeat step 5 onwards.

This is used throughout the Leapfrog algorithm to find the pseudo-fermion fields ϕ from known R and Q using the relation $Q\phi = R$, which requires the computation of Q^{-1} .

8.5 Simulation Plots, Observations and Inferences

Periodic boundary conditions have been imposed on both, the pseudo-fermion fields and the gauge fields. This cause the 2-D lattice to have a toriodal topology. $\hat{\mu}$ is taken as the spacial direction and $\hat{\nu}$ is taken as the temporal direction, Also, m is taken as the index for the spacial direction and n for the temporal direction.

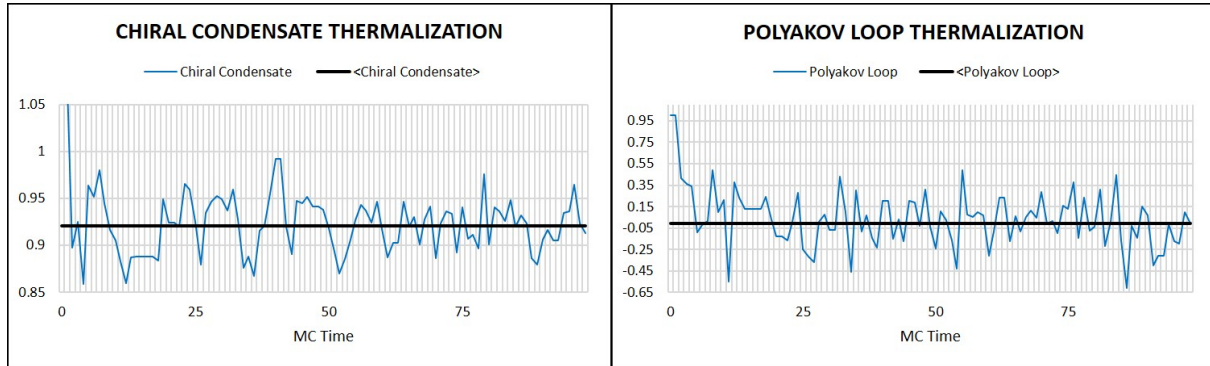
8.5.1 Simulation Parameters

- Coupling Constant (β): Ranges from 0.2000 to 6.0000
- $r = 1.0$ (Mass-like term)
- Fermion Mass (FM): $\{ 0.1, 0.2, 0.5 \}$
- Lattice size ($L_1 \times L_2$): 8×8
- HMC step size ($\Delta\tau$): 0.1000
- Number of HMC steps in between momentum updates: 10
- Thermalization steps (N_{Therm}): 100
- Number of measurements: 200
- MC updates between measurements: 5
- Fields were initialized by coldstart(): All gauge fields were initialized to 0 at the beginning of the simulation.

8.5.2 Thermalization of observables

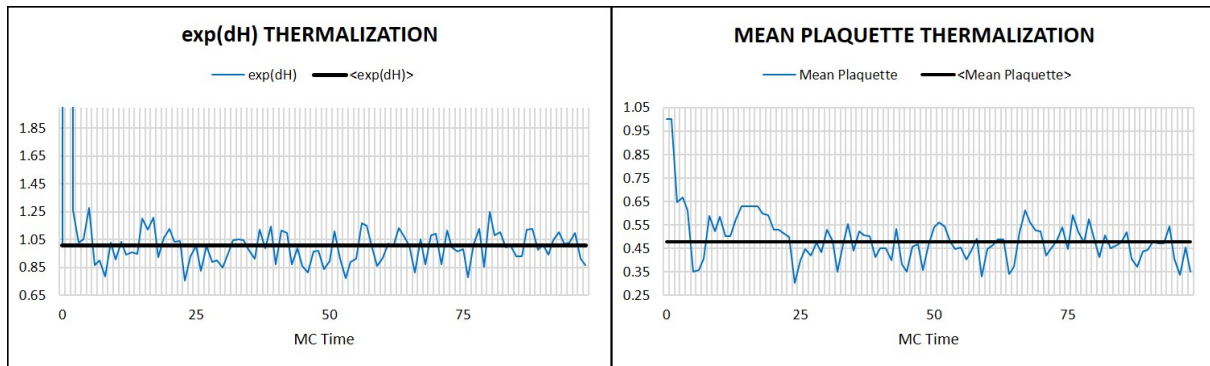
Ideally $N_{Therm} \rightarrow \infty$, as it takes infinite time for the HMC algorithm to equilibriate perfectly around the average. But since this is unphysical, a cutoff is defined at some finite N_{Therm} , and measurements of observables is done only for $N > N_{Therm}$.

There is no perfectly sound tool to calculate a good cutoff, since the value is different for different observables and simulation parameters. Hence the graphs of observables are plotted against MC time, and N_{Therm} is estimated through observations.



(a) Chiral condensate vs MC Time

(b) Polyakov loop vs MC Time



(c) Chiral condensate vs MC Time

(d) Polyakov loop vs MC Time

Figure 8.1: The thermalization of observables in lattice Schwinger model

From these plots, we observe that $N_{Therm} = 100$ is a good choice as the observables roughly start equilibrating around their respective average values.

8.5.3 Conjugate-gradient algorithm iterations

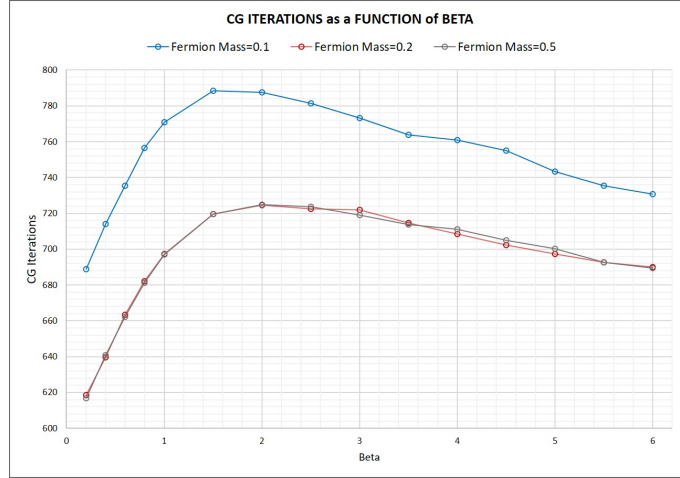


Figure 8.2: Mean number of iterations of the CG algorithm per HMC update as a function of β for different FM .

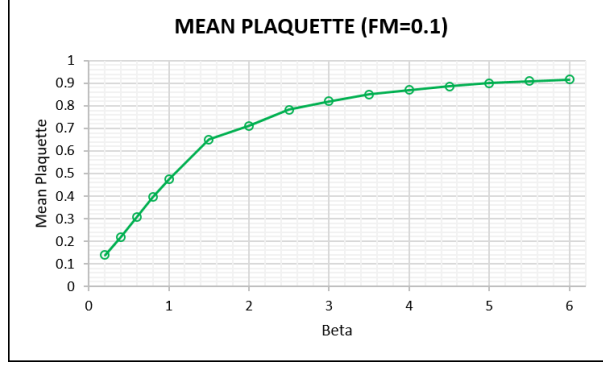
It is observed that the CG algorithm slows down as the fermions become lighter. The trend with respect to β is that it first increases as β increases, reaches a maximum and then decreases as β increases.

8.5.4 Mean plaquette (mp)

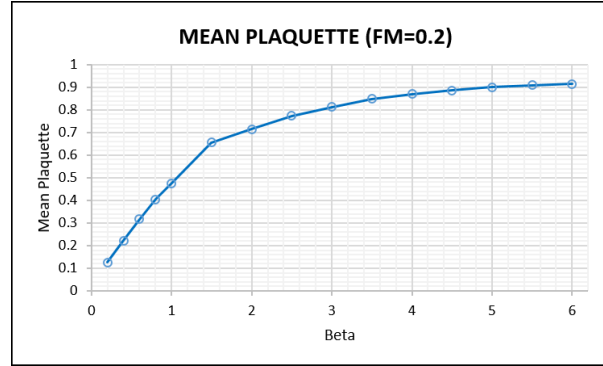
It is calculated by averaging the real part of the plaquette variable, $U_P = U_\mu(n)U_\nu(n + \hat{\mu})U_\mu^\dagger(n)U_\nu^\dagger(n + \hat{\mu})$ over all loops in the entire lattice.

$$\text{mp} = \sum_{\text{lattice}} \frac{\text{Re}[U_P]}{L_1 * L_2}, \quad (8.17)$$

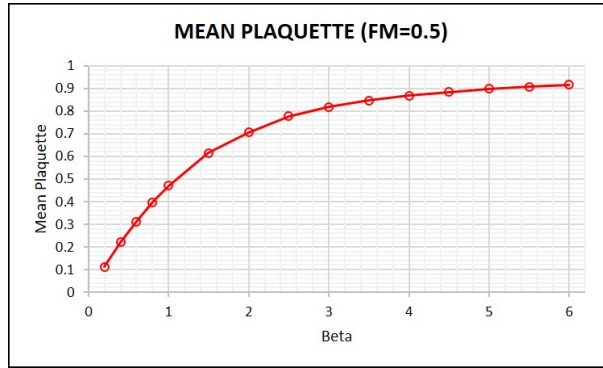
where L_1 and L_2 are the ultraviolet cutoffs of the lattice in directions 1 and 2.



(a) Fermion mass = 0.1



(b) Fermion mass = 0.2



(c) Fermion mass = 0.5

Figure 8.3: Mean plaquette as a function of β for different FM

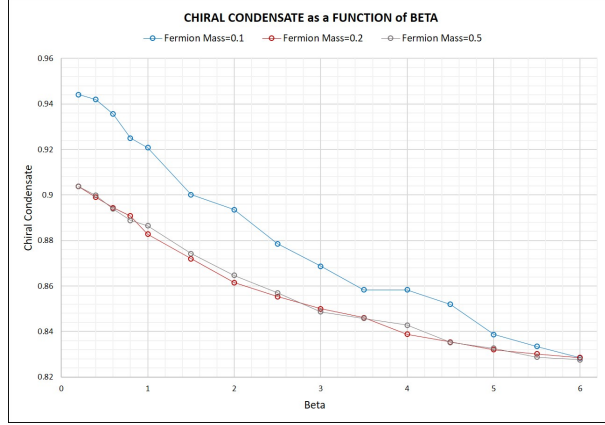
It is observed that the mean plaquette increases for increasing β , but the rate of increase decreases for increase in β . It is also observed that the mean plaquette is independent of the fermion mass.

8.5.5 Chiral condensate (cc)

It is calculated using the formula, where M is the Wilson Dirac operator and $\gamma_5 = \sigma_3$.

$$cc = Tr[\gamma_5 M^{-1}]. \quad (8.18)$$

A non zero value of chiral condensate implies that the fermions in this model are massive whereas a zero chiral condensate implies massless fermions.

Figure 8.4: Chiral condensate as a function of β for different FM .

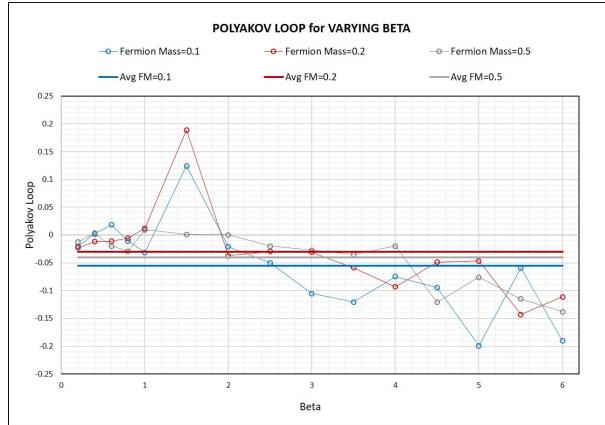
This model exhibits a spontaneous symmetry breaking of the $U(1)$ symmetry due to the presence of a non-zero chiral condensate, which in turn is produced through a pool of instantons. A non zero cc implies that the fermions are massive. But in this theory, since a mass-like term appears in the Wilson-Dirac operator (Eq 8.5), cc is still non-zero when the fermion mass is tuned to 0.

8.5.6 Polyakov loop (pl)

In gauge theory, a Wilson loop is a gauge-invariant observable obtained from the holonomy of the gauge connection around a given loop. The action of Wilson loop operators has the interpretation of creating an elementary excitation of the quantum field which is localized on the loop. The Polyakov loop is a Wilson loop along the temporal direction. The lattice has toroidal topology due to the imposed periodic boundary conditions in both directions. To visualize what the Polyakov loop is, we roll the lattice along the temporal direction and join the ends in the spacial direction. The Polyakov loop is product of U_ν along the ring in the temporal direction and this product is summed over the spacial direction. We have

$$pl = \sum_m \left[\prod_n U_\nu(n) \right]. \quad (8.19)$$

$pl = 0$ implies confined phase and $pl = 1$ implies deconfined phase in the theory.

Figure 8.5: Polyakov loop as a function of β for different FM .

This model exhibits confinement of the fermions inferred from the zero Polyakov loop. A possible argument on why this is so is because in two dimensions, classically, the potential between two charged particles goes linearly as r analogous to how to quark potential goes in the confined phase

$$V_q = A + \sigma r - \frac{B}{r}. \quad (8.20)$$

8.5.7 $\langle \exp(d\mathcal{H}) \rangle$

The introduction of conjugate momentum variables to the fields is solely to implement the HMC algorithm, this acts as a check to make sure that the algorithm is functioning well. $\langle \exp(d\mathcal{H}) \rangle \sim 1$ under experimental errors of the order of $(\Delta\tau)^2$.

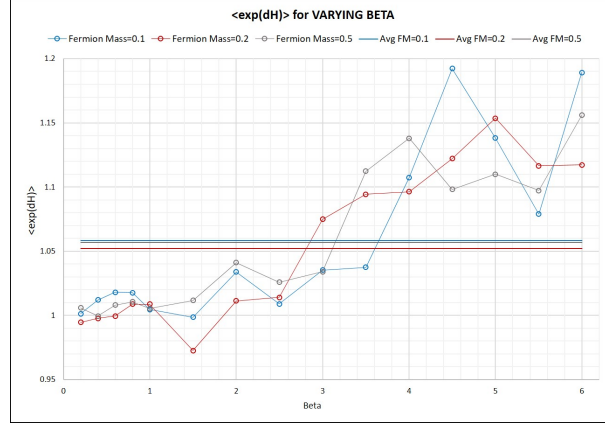


Figure 8.6: Mean of $\exp(d\mathcal{H})$ during the MD chain as a function of β for different FM .

It is observed that $\langle \exp(d\mathcal{H}) \rangle \sim 1$ under experimental errors, and hence our HMC algorithm works for these parameters.

9 Conclusions

The Schwinger Model was successfully simulated on a (1+1) dimensional lattice through the use of Hamiltonian Monte Carlo algorithm.

The expectation values for all observables were calculated after the Markov chain of the gauge and fermion field configurations on the lattice thermalized. The following conclusions were made from the plots:

- mean plaquette increases for increasing β , but the rate of increase decreases for increase in β . It is also observed that the mean plaquette is independent of the fermion mass.
- this model exhibits a spontaneous symmetry breaking of the $U(1)$ symmetry due to the presence of a non-zero chiral condensate. It also implies that the fermions are massive.
- this model exhibits confinement of the fermions inferred from the vanishing expectation value of the Polyakov loop. A possible argument on why this is so is because in two dimensions, classically, the potential between two charged particles goes linearly as r .

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